ResNet18

Bottle crack patch classification report

Run: 20261003_155746

Labels: good = 0; crack = 1.

Existing train/valid/test assignments were retained. Exact duplicate image pixels across splits were checked. This does not verify physical-bottle separation.

Split	Good images	Crack images
train	315	70
valid	105	14
test	32	8

Training procedure

ImageNet-pretrained ResNet18 with a new two-class output layer. All layers were fine-tuned using AdamW.

Input: RGB, full-patch resize to 224x224, ImageNet normalization. Training augmentation: horizontal flip.

Learning rate: 0.0001; weight decay: 0.0001; batch size: 16; seed: 42.

Completed epochs: 35; best epoch: 20; maximum epochs: 100; patience: 15; minimum significant loss improvement: 0.0001.

The best checkpoint was selected by validation loss. The threshold was then selected using validation crack F1. Test data were not used for model or threshold selection.

Selected threshold: 0.50

Unlike the HOG models, this experiment uses pretrained features, RGB 224x224 inputs, and training augmentation. Record these differences when comparing results.

Evaluation metrics

Metric	Train	Valid	Test
Accuracy	1.0000	1.0000	1.0000
Balanced accuracy	1.0000	1.0000	1.0000
Crack precision	1.0000	1.0000	1.0000
Crack recall	1.0000	1.0000	1.0000
Crack F1	1.0000	1.0000	1.0000
Specificity	1.0000	1.0000	1.0000
ROC AUC	1.0000	1.0000	1.0000
Average precision (AP)	1.0000	1.0000	1.0000

Split	Log loss	TP	TN	FP	FN
train	0.0004	70	315	0	0
valid	0.0017	14	105	0	0
test	0.0006	8	32	0	0

Final training metrics use the best model in evaluation mode without augmentation. They differ from training-batch metrics in the epoch history.

Training scores are resubstitution scores. Validation scores are selection scores. Test scores are held-out.

Average precision summarizes the precision-recall curve; it is not trapezoidal PR area. These are patch metrics, not whole-bottle inspection results.

Runtime and model

Training time: 135.45 seconds.

Mean patch pipeline time: 10.52 ms over 40 test images.

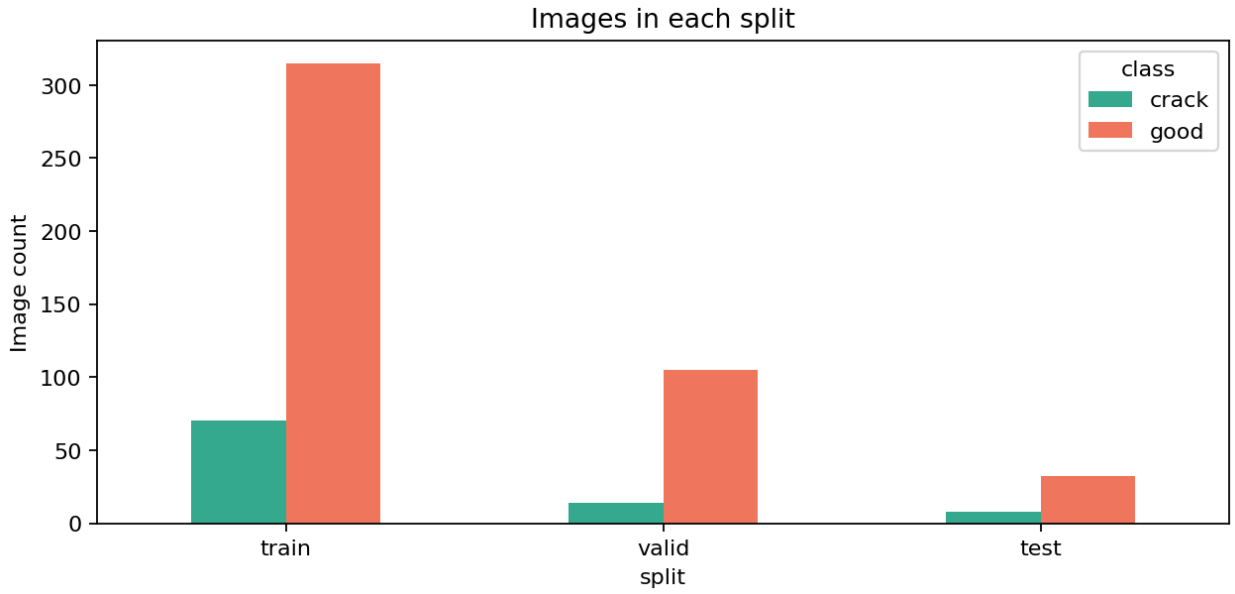
Timing includes image reading, preprocessing, device transfer and prediction. Warm-up is excluded. Hardware and Drive access affect these measurements.

Parameters: 11,177,538; model package size: 42.71 MB.

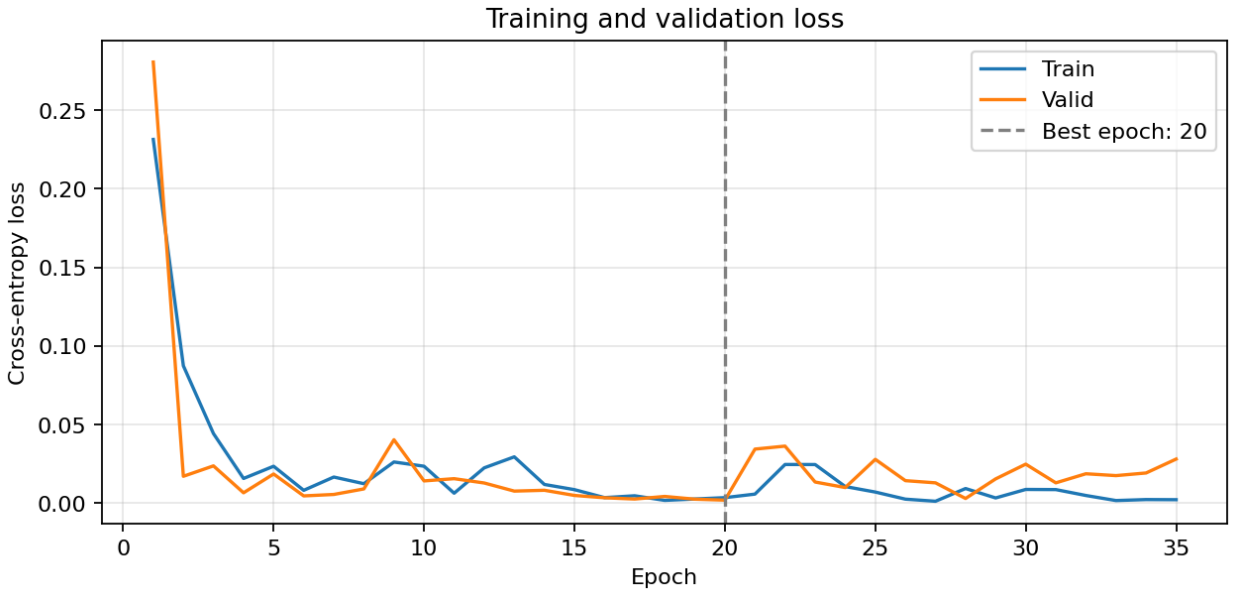
Hardware: Tesla T4

Python: 3.13.15; PyTorch: 2.11.0+cu130; Torchvision: 0.26.0+cu130.

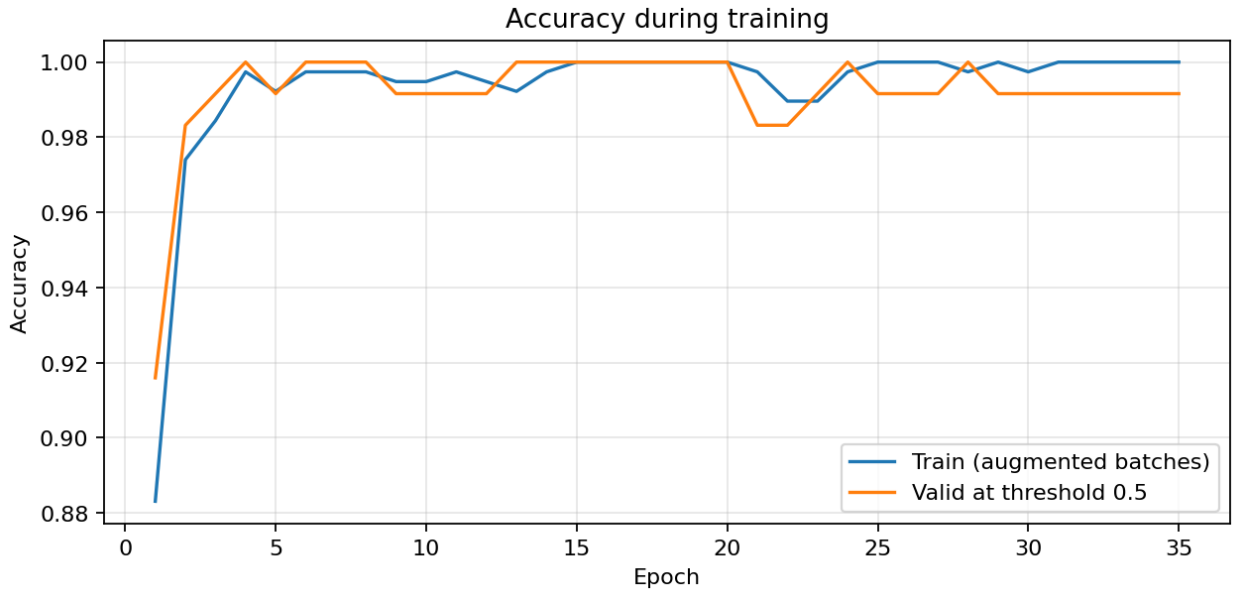
Dataset Distribution



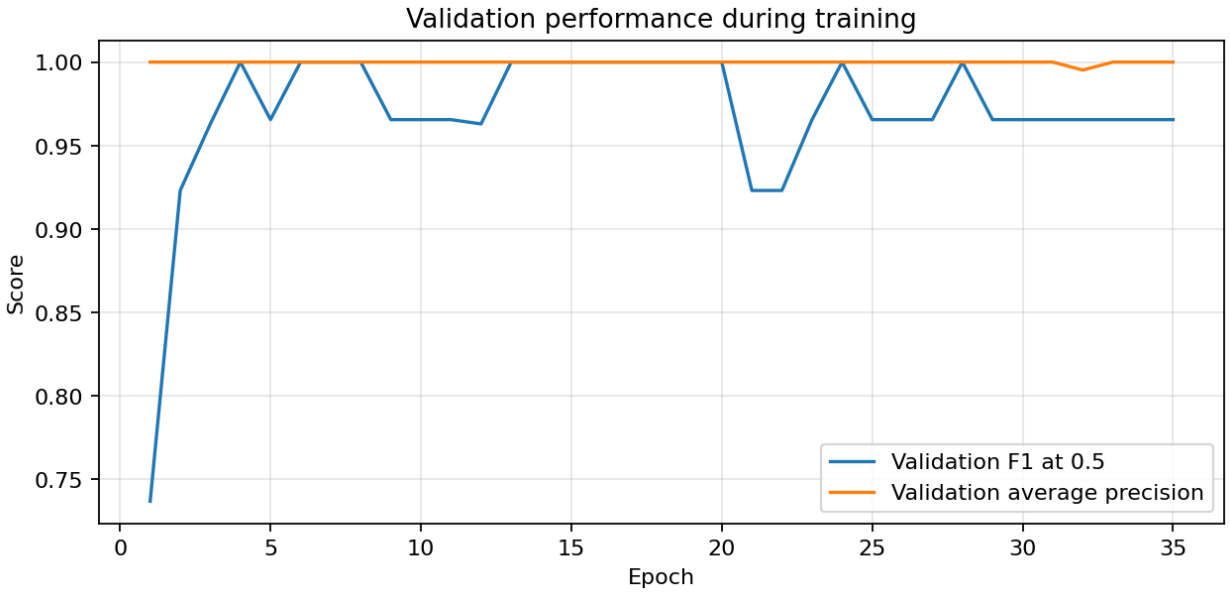
Training Loss



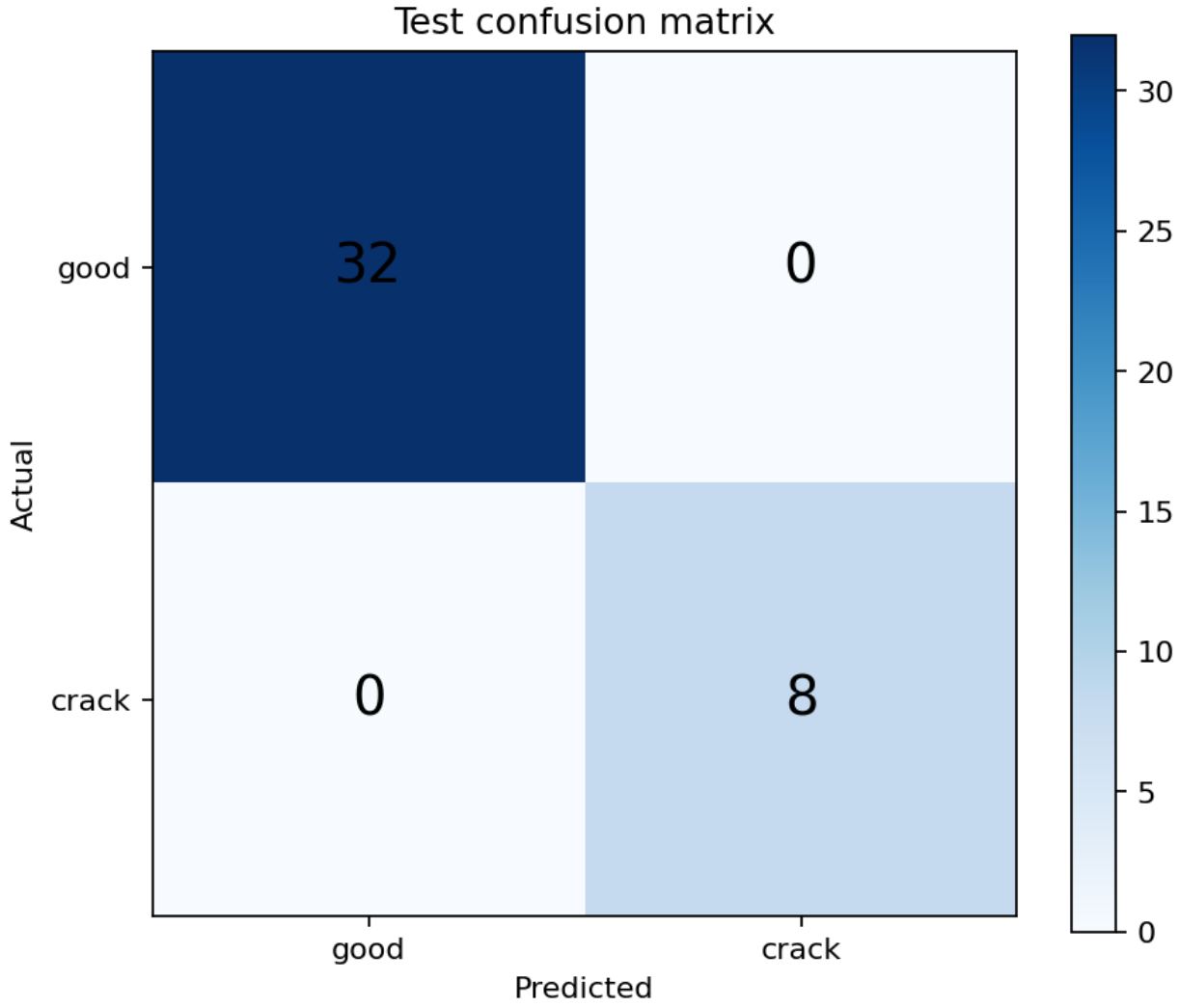
Training Accuracy



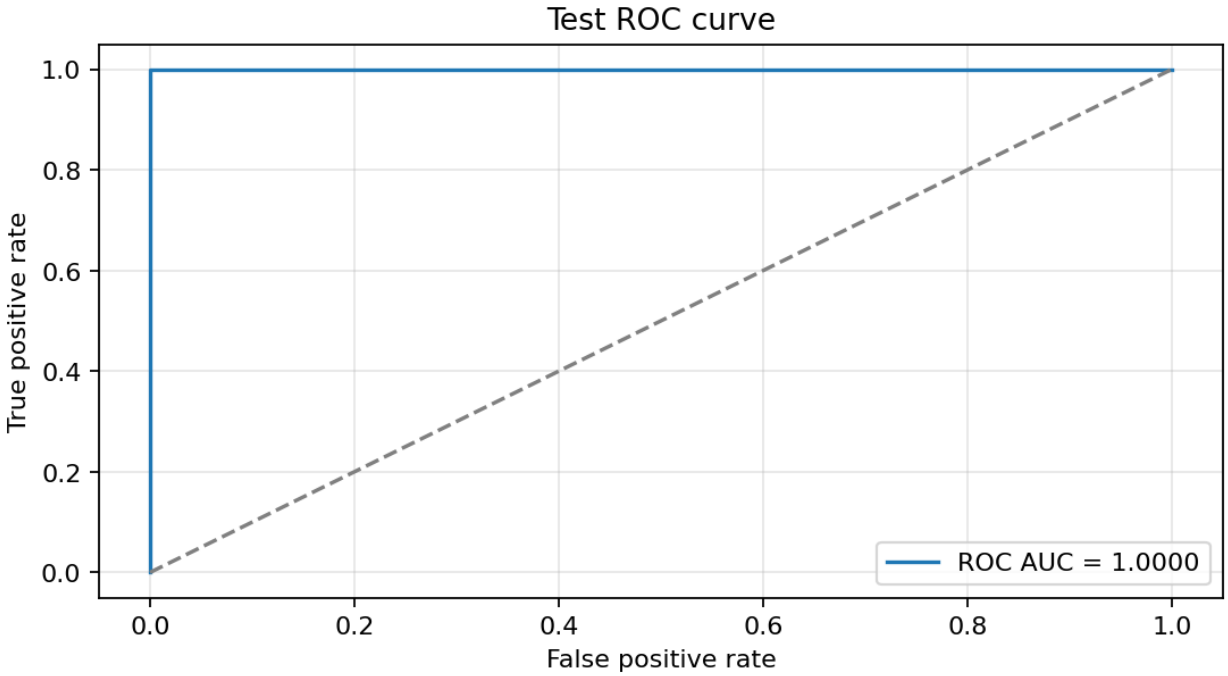
Validation History



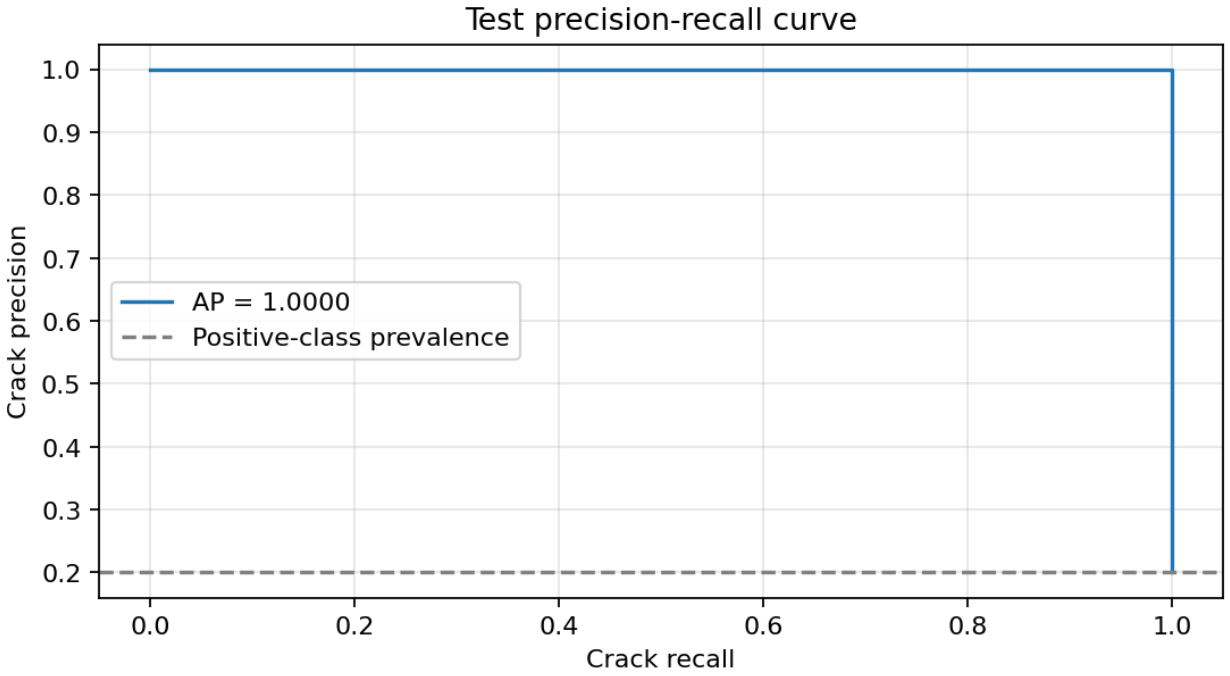
Test Confusion Matrix



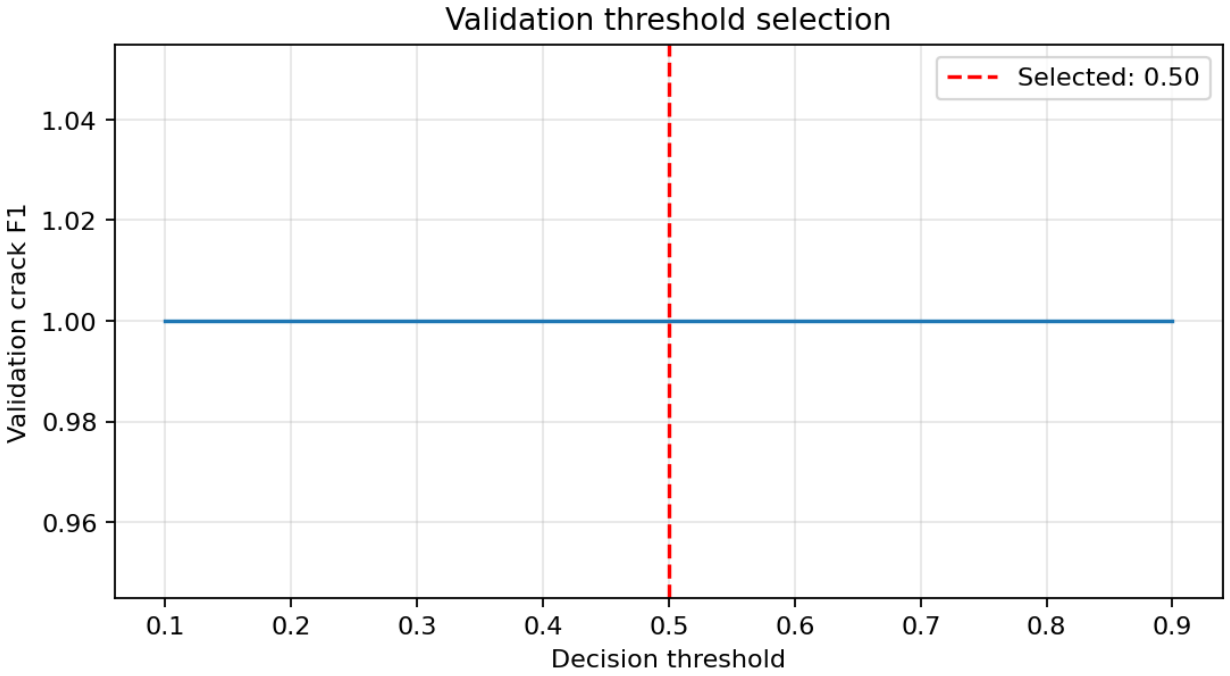
Test Roc


Test ROC curve
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC AUC = 1.0000

Test Precision Recall



Validation Threshold



Test Metrics

